Worksheet 6-6 A Properties of Integrals #1
AP Calculus Name_____

Given: $\int_{-3}^{2} f(x)\,dx = -4$ $\int_{2}^{7} f(x)\,dx = 7$ $\int_{-3}^{2} g(x)\,dx = 2$ $\int_{2}^{7} g(x)\,dx = 5$

**Directions:** Evaluate the following integral expressions, if they exist.

1.) $\int_{-3}^{2} -f(x)\,dx$

2.) $\int_{1}^{1} g(x)\,dx$

3.) $\int_{-3}^{7} f(x)\,dx$

4.) $\int_{2}^{7} [f(x) + g(x)]\,dx$

5.) $\int_{-3}^{2} [g(x) - f(x)]\,dx$

6.) $\int_{-3}^{2} 4\,dx$

7.) $\int_{7}^{2} f(x)\,dx$

8.) $\int_{-3}^{7} -g(x)\,dx$

9.) $\int_{-3}^{2} 7g(x)\,dx$

10.) $\int_{7}^{7} f(x)\,dx$

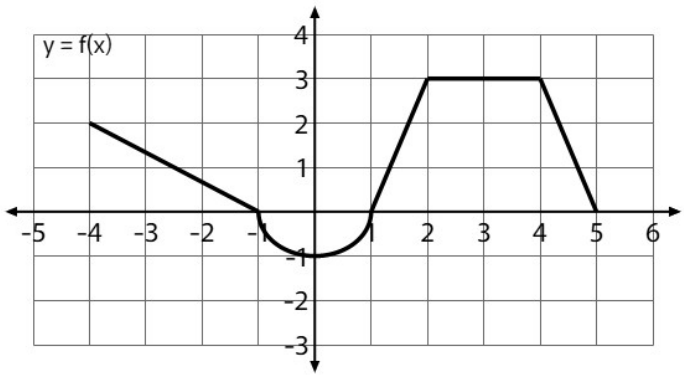
11.) $\int_{2}^{7} -2\,dx$

12.) $\int_{2}^{-3} g(x)\,dx$

13.) $\int_{-3}^{7} [2f(x) - 3g(x)]\,dx$

14.) $\int_{7}^{2} [-2f(x) + 6]\,dx$

**Directions:** The graph of $f$ consists of four line segments and a semicircle. Evaluate each definite integral.




15.) $\int_{-4}^{-1} f(x)\,dx$

16.) $\int_{-4}^{5} f(x)\,dx$

17.) $\int_{2}^{1} f(x)\,dx$

18.) $\int_{4}^{2} f(x)\,dx$

19.) $\left|\int_{-4}^{1} f(x)\,dx\right|$

20.) $\int_{-4}^{1} |f(x)|\,dx$

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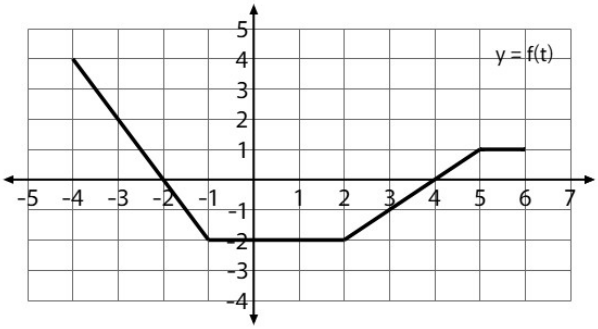
21.) The graph of the function $f$ is shown at the right. Let $g$ be the function defined by $g(x) = \int_2^x f(t)dt$.




(a) Find the average rate of change of $g$ from $x = -4$ to $x = 6$.

(b) Find the instantaneous rate of change of $g$ with respect to $x$ at $x = 5$, or state that it does not exist.

(c) On what open intervals, if any, is the graph of $g$ concave down? Justify your answer.